

Classification & reliability

ARN

Classification of Electronic Product

Parameter	Consumer Product	Industry Product	Military Product
Cost	Should be affordable	Development cost higher	Very High
Reliability	Good	Higher: Service @ customer sight	Highly reliable and durable
Ergonomics	Comfort and safety is important	Operator safety is important	Operator safety is important
Aesthetics	Attractive	Look of product is not important	Look of product is not important
Life	Moderate	Better	Best
Maintenance	Minimum	Minimum	Lowest
Operating temp. range	0 to 70 °C	-25 to 85 °C	-55 to 125 °C

Ergonomics

- Efficient and effective interconnect between man machine and environment around it.
- Ergonomic design should
 - reduce user fatigue, stress,
 - improve safety, comfort, job satisfaction, quality of life

Reliability Engineering

- Ability of product to perform intended function under stated condition, for stated period of time
- It is probability the unit perform intended functionality for stated period of time for stated operating conditions

Reliability of equipment

- Design of equipment
- Quality of Manufacturing
- Storage of product
- Environmental conditions

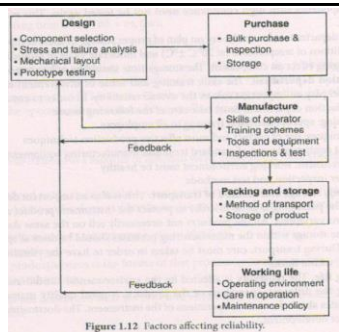


Figure 1.12 Factors affecting reliability.

- (a) **Design stage.** While designing and selecting the components, the targeted figure of reliability should be decided. In doing so, the designer should keep in mind the following points:
- Selection of component must be done properly.
 - No undue stresses should act on component.
 - Investigation of individual component, or sample check, is must.
 - The circuit can be tested by available CAD software.
 - Since the mechanical layout of components, assemblies and panels is very important, components should not be mechanically stressed by poor mounting.
 - Heat developed within the system should be estimated and maintained within the limits.
 - Based on the requirements suitable protection must be added.
 - The appropriate environment testing—hermetic sealing, forced air cooling, etc.—should be carried out.
 - Extensive prototype testing should be carried out to meet the performance and reliability specifications.

- Purchase department.** The purchase department should work out on the following points:
- Ordering the required quantity of components to reputed vendor.
 - Obtaining the components at the right time and at the low price, for example, the correct purchasing policies.
 - The component must be checked by the inward inspection section. They may test/inspect all the components, or a sample check can be carried out. If the components are available from a reputed manufacturer, then depending on the confidence level, the sample testing can be done.
 - Components with high confidence need not be tested again. This will definitely save the time.
 - This department also works out on plan of proper storage of these components at the specific conditions of temperature (at $20^{\circ}\text{C} \pm 2^{\circ}\text{C}$) and moderate value of humidity in order to reduce the aging effect on component. The storage time should be as less as possible.

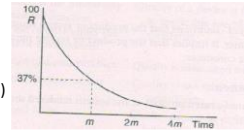
- Production department.** The skill, training and sense of involvement of all the production personnel play an important role in the overall reliability. In order to ensure a good reliability, the production department must take care of the following issues:
- Training: appropriate/latest training to all employees
 - Techniques: use of correct and most effective production techniques
 - Latest equipments: use of standard tools and manufacturing equipments
 - Environment: working environment must be healthy
 - Proper inspection and test methods

Packaging, storage and method of transport. This is also an important deciding factor of reliability. The packaging is done in order to protect the instrument/product against environment conditions variations. All the products not necessarily sell on the same day of manufacturing. Hence the storage within the manufacturing premises should be done at specified storage conditions. During transport, care must be taken in order to have the vibration, shock, temperature, etc., within limits.

Working life. The reliability is affected by the environmental conditions and correctness of handling the different operations of the product. A good quality manual will provide the detailed idea about the various operations of the instrument. The documentation is integral part of product development.

Exponential law of reliability

- Exponential law of reliability
- $R = e^{-\lambda t}$
- R = reliability of Equipment
- λ = system failure rate (Failure per month)
- If $m = 1/\lambda$
- Then $R = e^{-t/m}$ This is an exponential law of reliability
- If $t = m$ then $R = e^{-1} = 0.37$
 - I.e. if $t = m$ then probability of successful operation reduces to 37 %



Failure or Defect

- Component or system may fail after some time there by reliability is always less than 100%

Types of failure	Phase at which failure occurs
Initial	During initial phase
Early	Immediately after short time of installation
Random	During arbitrary instance
Catastrophic	Major failure for long time
Wear out	After long usage of equipment

Factors responsible for failure

- Incorrect design
- Wrong manufacturing process
- Complexity of equipment
- Poor quality control
- Improper storage or transportation
- Misuse/ mishandling of equipment
- Human error

Maintainability

- It is probability that failed equipment is restored to meet designed specifications in known time called down time when maintenance performed under stated conditions
- Maintainability is the probability that the failed equipment will be repaired within time t , and T is random variable which represents the repair time then Maintainability is given by
 - $M(t) = \Pr(T \leq t)$;
 - $M(t)$ = Maintainability with time
 - Where $M(t) = \Pr(T \leq t)$,
 - $\Pr(t)$ repairing probability within time T

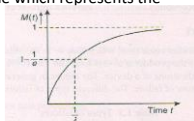


Figure 1.14 Graph of maintainability vs. time.

Bath tub curve

- The failure of an item can be grouped into three different modes, depending on the nature of the failure. The typical curve which indicates these modes of the failure or behavior is shown in fig having shape similar to bathtub
- The failure rate of components
 - Can be found by operating components for long duration and observing number of failures

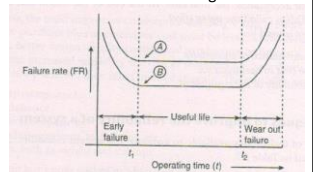


Figure 1.16 Bathtub curve.

Early failure reasons

- Defect or errors during manufacturing
- Poor quality raw material
- Wrong assembly
- Improper insulation
- Poor fitting
- Early failure period is also called as burn-in-period

Random failure

- Failure rate is constant
- Mishandling/ Exceeding operating condition damage components

Wear out failure

Wear-out failure

After the period of the useful life, the units get worn-out and begin to deteriorate, and the failure rate goes on increasing. This period is called the *wear-out region* (see Figure 1.16). In this period, there is a gradual change in the values of the parameter, determining performance, and when parameters go beyond the limit of acceptance, the unit fails. At the end of the useful life, the item is replaced by another tested item. This is called the *aging effect*.

The bath tub curve is used to:

- Decide the maintenance of equipment
- Determine the cost of maintenance
- Predict the useful life of the equipment

- The number of components are kept in operation at maximum rated condition
- Failure rate to time is observed and plotted as failure rate V/s time
- The graph can be plotted for two different stress severity
- Graph A = High stress severity
- Graph B = Low stress severity

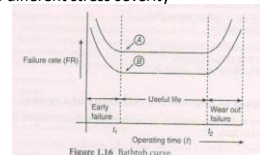


Figure 1.16 Bathtub curve.